

# Baseline Configuration and Random-Sampling Controls in Metaheuristic Comparisons for Wind Farm Layout Optimization

Prince Solanki, Pulkit Dwivedi, Vanita Garg, and Vinay Shukla

**Abstract**—We compare eight wind farm layout optimizers on the discretized Kusiak–Song benchmark (68 cases,  $15^\circ$  rose) with 6,030 evaluations from random starts, feasibility-aware ranking and equivalence tests. First, the particle swarm optimization (PSO) setting  $w = 0.7$ ,  $c_1 = c_2 = 2$  lies beyond the order-2 (variance) convergence boundary; in a sweep of  $c_1 = c_2$ , feasibility starts to degrade there and collapses with larger steps. With it, PSO ranks behind two salp-swarm hybrids in the eight-method pool of an earlier comparison; with the constriction setting, it ranks first there. Second, a salp-swarm (SSA) first phase adds nothing beyond equal-cost random sampling in the farm (a gain above 0.05 percentage points (pp) of wake loss is ruled out across seeds and cases), the Laplacian SSA as published is worse, and a PSO phase gives a clear gain; at this budget the second phase, variable neighborhood search (VNS), acts essentially as one compass-search descent. Third, PSO followed by VNS (PSO-VNS) has the best average rank with  $15^\circ$ ,  $5^\circ$  and  $1^\circ$  direction bins, also under a Gaussian wake at  $1^\circ$ , but its average advantage over a well-configured PSO is small: within  $\pm 0.05$  pp across seeds and cases, not farm clusters, under the  $15^\circ$  rose, and a significant  $-0.043$  pp under a  $1^\circ$  rose; case-level differences go both ways. PSO-VNS reaches feasibility more reliably (Horns Rev 1 block: 30/30 vs 19/30 runs, McNemar  $p = 9.8 \times 10^{-4}$ ).

**Index Terms**—Benchmarking, equivalence testing, particle swarm optimization, random-sampling control, wake effect, wind farm layout optimization.

## NOMENCLATURE

|                    |                                                     |
|--------------------|-----------------------------------------------------|
| $B$                | Budget (number of evaluations).                     |
| $C_T, K$           | Thrust coefficient; Jensen wake-spreading constant. |
| $D, R$             | Rotor diameter and rotor radius ( $D = 2R$ ).       |
| $E_{\text{ideal}}$ | Wake-free benchmark objective of one turbine.       |
| $F_p$              | Penalized wake loss (minimized).                    |
| $\ell_{\min}$      | Minimum spacing between turbines.                   |

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P. Solanki [TBD: mark corresponding author] is with the Department of Mathematics, School of Sciences, IILM University, Greater Noida, Uttar Pradesh, India; he was previously with the Department of Mathematics, Indian Institute of Technology Roorkee, Roorkee, India (e-mail: prince@ma.iitr.ac.in; prince@iilm.edu).

P. Dwivedi is with the Department of Computer Science and Engineering, Bennett University, Greater Noida, Uttar Pradesh, India (e-mail: pdwivedi19900@gmail.com; pulkit.dwivedi@bennett.edu.in).

V. Garg is with Amity University, Noida, Uttar Pradesh, India (e-mail: vanitagarg16@gmail.com).

V. Shukla is with the School of Mathematical Sciences, Shanghai Jiao Tong University, Shanghai 200240, China, and also with the Department of Mathematics, School of Computer Science Engineering and Technology, Bennett University, Greater Noida, India (e-mail: vinayshukla4321@gmail.com; vnsukla01@sjtu.edu.cn).

|                 |                                                      |
|-----------------|------------------------------------------------------|
| $N, N_p$        | Number of turbines; population size.                 |
| $N_\theta, N_s$ | Numbers of wind-direction and wind-speed bins.       |
| $r$             | Farm radius.                                         |
| $w, c_1, c_2$   | Inertia weight and acceleration coefficients of PSO. |
| $\mathbf{X}$    | Layout vector $(x_1, y_1, \dots, x_N, y_N)$ .        |
| $\theta$        | Wind direction (blowing towards, CCW from east).     |
| $\omega$        | Fraction of the budget given to Phase 1.             |

## I. INTRODUCTION

The positions of the turbines in a wind farm are fixed at the planning stage and determine, for its lifetime, how much of the available wind energy it converts. An upstream turbine leaves behind a wake of slower and more turbulent air that reduces the power of the turbines downstream, so the layout sets the wake loss and thereby the gross energy yield, and shapes the collector system and grid connection [1], [2]. As the wake loss is a share of the wake-free energy, a difference of a few tenths of a percentage point is, within a given wake model, a lifetime energy difference of that size. The wind farm layout optimization problem (WFLOP) places a given number of turbines in a site to minimize the wake loss subject to boundary and minimum-spacing constraints; we address only this wake-loss part of layout design, for farms of up to 36 turbines.

Early studies placed turbines on a grid with genetic algorithms (GAs) [3], [4]. The continuous model of Kusiak and Song [5], a common reference for metaheuristics, has been solved with ant colony optimization and particle filtering [6], [7] and biogeography-based optimization [8]; GA, particle swarm optimization (PSO), differential evolution (DE) and simulated annealing were compared in [9], and PSO has reduced wake losses in [10]. Later work used informed GAs [11], [12], deterministic refinement [13], variable neighborhood search (VNS) [14], pseudo-gradients [15], Gaussian wake models [16] and surrogates [17]–[19].

Guidelines for comparing metaheuristics are well established: controlled rather than competitive experiments [20], equal budgets, reported settings, component analysis and paired nonparametric tests with multiplicity correction [21]–[24], and equivalence or Bayesian analyses of practically relevant differences [25], [26]. Several metaphor-based algorithms, the salp swarm algorithm (SSA) [27] among them, reuse

known search operators [28]–[30]. In the WFLOP, the GECCO layout competition [31] and the IEA Wind Task 37 studies [32], [33] compared optimizers on a common objective, and reviews list many metaheuristics tested under differing budgets and settings [34]; these studies compare methods as submitted or as published, not how the configuration of a baseline or the choice of control changes which method appears best. Four issues matter here. First, the PSO setting  $w = 0.7$ ,  $c_1 = c_2 = 2$  (inertia weight and acceleration coefficients), which keeps the acceleration coefficients of the original PSO [35] and was used in our own earlier comparison, lies in the order-1 (mean) convergence region but outside the smaller order-2 (variance) region [36]–[38]. Second, budgets counted in iterations, or without the gradient evaluations of gradient-based solvers, are not equal. Third, infeasible layouts returned by penalty-based metaheuristics must be ranked consistently. Fourth, mean values without paired tests and multiplicity correction [39] may mislead, and a nonsignificant difference is not evidence of equivalence [26]. For two-phase hybrids, a further question is whether the first phase contributes more than a random start would.

Our thesis is that the conclusions of such comparisons depend strongly on how the baselines are configured and how the comparison is controlled. Under a controlled protocol at 6,030 evaluations from random starts, a simple two-phase design, PSO followed by the basic VNS [40], [41] (PSO-VNS), and a well-configured PSO rank first and second of eight methods and differ little on average, whereas salp-swarm first phases add nothing beyond random sampling in the farm and the Laplacian SSA (LX-SSA) [42], proposed earlier by two of the authors, is worse as published. PSO-VNS is a sequential hybrid in the memetic sense, not a new PSO variant. Our contributions are:

- 1) *Protocol*. Established practice [22]–[24] applied to the continuous WFLOP (equal budgets counting gradient evaluations, seed-paired runs, Holm-corrected tests, public code and data), with three elements not reported for this problem to our knowledge: feasibility-aware ranking, equal-cost random-sampling controls for hybrids, and an equivalence analysis [25], [26] at the seed, case and cluster level with a post hoc margin of  $\pm 0.05$  percentage points (pp) of wake loss; applied to 68 benchmark cases, a Horns Rev 1 block and IEA37 Case Study 1 (Sections V and VIII).
- 2) *Baseline configuration*. The PSO setting  $w = 0.7$ ,  $c_1 = c_2 = 2$  (run without velocity clamping) lies beyond the order-2 boundary (Proposition 1); in a sweep of  $c_1 = c_2$ , feasibility starts to degrade at the boundary and collapses with larger steps, and the setting reverses the order of PSO and the salp-swarm hybrids (Section VI-A).
- 3) *Component analysis*. Against random sampling in the farm (RSD-VNS), salp-swarm first phases add nothing and LX-SSA as published (at twice the evaluations per iteration) is worse, whereas a PSO phase gives a clear gain; at 6,030 evaluations the VNS phase acts essentially as one compass-search descent (Section VII).
- 4) *PSO-VNS versus PSO*. PSO-VNS has the best average

rank at every direction resolution tried, but its average advantage over a well-configured PSO is small (equivalent at  $\pm 0.05$  pp across seeds and cases, not clusters, with the  $15^\circ$  rose;  $-0.043$  pp, significant, with a  $1^\circ$  rose), and case-level differences go both ways. We report where it gains (Horns Rev 1 feasibility, larger Data Set II layouts) and where it does not (feasible starts, IEA37), and derive recommendations (Sections VI-B–X).

## II. PROBLEM MODELING

For a prescribed number  $N$  of identical turbines, the  $n = 2N$  coordinates  $(x_i, y_i)$  are optimized to maximize the expected farm power, i.e., to minimize the wake loss, under the Jensen wake model [43], [44] and the power curve of [5]. The wind direction  $\theta$  has the probability density  $p_\theta$ , and for each direction the wind speed  $s$  follows a Weibull density  $p_s(s; \zeta, \psi)$  with shape  $\zeta(\theta)$  and scale  $\psi(\theta)$ . Here  $\theta$  is the direction *towards* which the wind blows, counter-clockwise from east; the meteorological directions  $\theta_{\text{met}}$  of Horns Rev 1 and IEA37 (wind *from*, clockwise from north) are converted by  $\theta = 270^\circ - \theta_{\text{met}}$ . Two constraints apply: all turbine pairs satisfy  $\ell_{ij} = \|(x_i - x_j, y_i - y_j)\|_2 \geq \ell_{\min} = 4D = 8R$ , as in [5], [8], and the farm is circular,  $x_i^2 + y_i^2 \leq r^2$ , with  $r = 500, 750$  or  $1000$  m.

### A. Wake Effect Model

In the Jensen model (Figs. S17–S19), the wake of turbine  $j$  is a half cone with half-angle  $\alpha = \arctan K$  (wake-spreading constant  $K$ ), and its virtual vertex lies  $R/K$  upstream of the rotor. With  $u_{ij} = (x_i - x_j) \cos \theta + (y_i - y_j) \sin \theta + R/K$  and the lateral offset  $v_{ij} = -(x_i - x_j) \sin \theta + (y_i - y_j) \cos \theta$ , turbine  $i$  lies in the wake if  $\beta_{ij} = \cos^{-1}(u_{ij} / \sqrt{u_{ij}^2 + v_{ij}^2}) < \alpha$ . Its downstream distance is  $d_{ij} = |u_{ij} - R/K|$ , and with the thrust coefficient  $C_T$  the velocity deficit is

$$\Delta_{ij} = 1 - \frac{s_{\text{down}}}{s_{\text{up}}} = \frac{1 - \sqrt{1 - C_T}}{(1 + K d_{ij}/R)^2}. \quad (1)$$

As in [5], this top-hat wake is tested at the hub center only, so the deficit jumps at the cone edge, and it also reaches a turbine less than  $R/K$  *upstream* of  $j$  near its axis (non-physical, kept for comparability). The deficits are combined by the root-sum-square rule,

$$\Delta_i(\theta) = \left( \sum_{j \neq i, \beta_{ij} < \alpha} \Delta_{ij}^2 \right)^{1/2}. \quad (2)$$

As  $C_T$  is constant (which overstates the deficits above rated speed), the local wind speed in a direction bin is the free-stream speed times  $1 - \Delta_i(\theta)$ , a Weibull variable with shape  $\zeta(\theta)$  and scale  $\psi_i(\theta) = \psi(\theta)[1 - \Delta_i(\theta)]$ .

### B. Power Model

The power curve of the Kusiak–Song benchmark [5] is  $f(s) = 0$  for  $s < s_{\text{cut-in}}$ ,  $f(s) = \lambda_1 s + \lambda_0$  for  $s_{\text{cut-in}} \leq s \leq s_{\text{rated}}$  and  $f(s) = P_{\text{rated}}$  above, with  $s_{\text{cut-in}} = 3.5$  m/s,  $s_{\text{rated}} = 14$  m/s,  $P_{\text{rated}} = 1500$  kW,  $\lambda_1 = 140.86$  and  $\lambda_0 = -500$ . We use it as published: it is slightly negative just above the cut-in speed,

jumps from 1472 to 1500 kW at the rated speed and has no cut-out speed. The expected power of turbine  $i$ ,

$$E(P_i) = \int_0^{360} p_\theta(\theta) \int_0^\infty f(s) p_s(s; \zeta(\theta), \psi_i(\theta)) ds d\theta, \quad (3)$$

is evaluated, as in [5], by a Riemann sum over  $N_\theta$  direction bins with probabilities  $p_l$  and  $N_s$  speed bins between  $s_{\text{cut-in}}$  and  $s_{\text{rated}}$  (Section S-VIII-A). The direction sum keeps the bin width in degrees as a factor; with 24 bins of  $15^\circ$  and  $\sum_l p_l = 1$ , the benchmark objective  $\sum_i E(P_i)$  is thus 15 times the expected farm power in kW, and the gross (wake-only) AEP of the benchmark is  $\text{AEP [MWh]} = (\text{objective}/15) \times 8.76$ . For Data Set I, the wake-free benchmark objective of one turbine,  $E_{\text{ideal}} = 14,045.74$ , corresponds to 936.38 kW (a capacity factor of 62%, far above real sites) or 8,202.7 MWh/yr.

### C. Constraint Handling

After every position update, each coordinate is clipped to  $[-r, r]$  (box repair; the square, not the circle). The constraints are handled by a penalty on the violations  $g_i^b = x_i^2 + y_i^2 - r^2$  and  $g_{ij}^s = \ell_{\min} - \ell_{ij}$ ; all optimizers except MS-SLSQP minimize the penalized wake loss

$$F_p(\mathbf{X}) = \left( NE_{\text{ideal}} - \sum_{i=1}^N E(P_i) \right) + \sum_{i: g_i^b > 0} (1 + \mu g_i^b)^2 + \sum_{i < j: g_{ij}^s > 0} (1 + \mu g_{ij}^s)^2, \quad \mu = 10^{10}, \quad (4)$$

where  $\mathbf{X} = (x_1, y_1, \dots, x_N, y_N)$ ; since  $NE_{\text{ideal}}$  is the wake-free benchmark objective, the first term is the wake loss reported in our tables. Any violation above about  $4.6 \cdot 10^{-8}$  (m or  $\text{m}^2$ ) makes the penalty exceed any possible wake loss, so comparing  $F_p$  values is, up to smaller violations, a feasibility-first rule (feasible before infeasible, then wake loss or total penalty). This penalty scaling ( $g_i^b$  in  $\text{m}^2$ ,  $g_{ij}^s$  in m) favours infeasible layouts inside the farm with turbines too close, or with a turbine clipped to the box near a tangent point  $(\pm r, y_i)$ , where  $g_i^b = y_i^2$  is second order in  $y_i$ . MS-SLSQP minimizes the wake loss under explicit constraints. A final layout counts as feasible if  $\ell_{ij} \geq \ell_{\min}$  and  $\sqrt{x_i^2 + y_i^2} \leq r$  hold within  $10^{-6}$  m (Section V), a looser tolerance than the penalty's.

## III. PRELIMINARIES: PHASE-1 METHODS AND BASIC VNS

A population method keeps  $N_p$  candidate vectors  $\mathbf{X}^i \in \mathbb{R}^n$  with components  $X_m^i$  and bounds  $lb_m, ub_m$ ; unless stated otherwise, one iteration costs  $N_p$  objective evaluations.

### A. PSO and Its Stability Conditions

We use the canonical global-best PSO [35] in the inertia-weight form [45]. Particle  $i$  has a position  $\mathbf{X}^i$ , a velocity  $\mathbf{V}^i$  and a personal best  $\mathbf{P}^i$ , the swarm keeps the global best  $\mathbf{G}$ , and a particle moves by

$$\mathbf{V}^i \leftarrow w \mathbf{V}^i + c_1 \rho_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + c_2 \rho_2 \circ (\mathbf{G} - \mathbf{X}^i), \quad (5)$$

$$\mathbf{X}^i \leftarrow \mathbf{X}^i + \mathbf{V}^i, \quad (6)$$

where  $\rho_1, \rho_2$  are vectors of independent  $U(0, 1)$  numbers, drawn anew for every move, and  $\circ$  is the componentwise product. The constriction PSO of Clerc and Kennedy [46] with their standard choice  $\varphi_1 = \varphi_2 = 2.05$  (constriction factor  $\kappa = 0.7298$ ) is identical to (5) with

$$w = 0.7298, \quad c_1 = c_2 = \kappa \varphi_1 = 1.49618. \quad (7)$$

PSO denotes (5), (6) with these coefficients, fixed a priori from the literature, and the *old setting* the same algorithm with  $w = 0.7$ ,  $c_1 = c_2 = 2$ , i.e., the acceleration coefficients of the original PSO [35] combined with an inertia weight. Both are run as implemented: velocities start at zero and are not clamped (no  $V_{\max}$ ), a coordinate clipped to the square (Section II-C) keeps its velocity, and  $\mathbf{P}^i, \mathbf{G}$  are updated after each evaluation.

*Proposition 1:* Under stagnation (fixed  $\mathbf{P}^i, \mathbf{G}$ ) and without active bounds, the expected position converges to  $(c_1 \mathbf{P}^i + c_2 \mathbf{G}) / (c_1 + c_2)$  for all initial values if and only if  $|w| < 1$  and  $0 < c_1 + c_2 < 4(1 + w)$  (order-1 stability;  $(c_1 + c_2)/2 = E(c_1 \rho_1 + c_2 \rho_2)$ ). For  $c_1 = c_2$ , the second moments converge as well (order-2 stability), with a limit variance  $\propto (P_m^i - G_m)^2$  in coordinate  $m$ , if and only if, in addition [36],

$$c_1 + c_2 < \frac{24(1 - w^2)}{7 - 5w}. \quad (8)$$

Proofs, also for  $c_1 \neq c_2$  (stricter bound): Section S-VI-A; for  $c_1 = c_2$ , the region also holds under a weaker, non-stagnant assumption [38]. The constriction setting is order-2 stable ( $c_1 + c_2 = 2.99 < 3.35$ ). The old setting is order-1 stable ( $4 < 6.8$ ) but lies beyond the order-2 boundary ( $4 > 3.50$ ;  $c_1 = c_2 > c^* = 1.7486$  at  $w = 0.7$ ; Table S50). In a sweep of  $c_1 = c_2 = c$  at  $w = 0.7$  (Fig. S1, Table S3),  $\geq 98.9\%$  of the final layouts are feasible up to  $c = 1.7$ ; beyond  $c^*$ , feasibility falls progressively (to 58.9% at  $c = 2$ ), so the boundary marks approximately where this PSO starts to fail. Clamping velocities to  $0.2(ub_m - lb_m)$  largely rescues the old setting (92.2% feasible), zeroing clipped ones does not (57.2%): it fails mainly through large steps. Its swarm contracts much less (Section VI-A); 224 of its 286 infeasible final layouts violate only the spacing, and 60 of the other 62 have a turbine pinned at the box near a tangent point (Section II-C). DE's  $F = 0.5$  exceeds Zaharie's critical value (0.14 at CR = 0.9) [47], so its weak results are not due to premature variance loss.

### B. Salp Swarm Algorithm and Its Laplacian Variant

The SSA [27] keeps the best solution found so far as the food source  $\mathbf{H}$  and splits the sorted population into leaders and followers. A leader  $i \leq N_p/2$  updates each coordinate by

$$X_m^i = H_m \pm \xi_1 ((ub_m - lb_m)\xi_2 + lb_m), \quad (9)$$

with either sign with probability  $1/2$ ,  $\xi_2 \sim U(0, 1)$  and  $\xi_1 = 2 \exp[-(4t/T)^2]$  decreasing with the iteration  $t$  out of  $T$ ; a follower  $i > N_p/2$  moves to the midpoint of its position and the new position of the preceding salp. The leaders thus perturb  $\mathbf{H}$  uniformly with a shrinking radius, and the followers only average [30]. LX-SSA [42] gives each follower a second candidate  $Y_m^i = X_m^i + \gamma(H_m - X_m^i)$  with the Laplace step [48]

$$\gamma = \phi - \chi \operatorname{sgn}(z - \tfrac{1}{2}) \ln(1 - 2|z - \tfrac{1}{2}|), \quad z \sim U(0, 1), \quad (10)$$

$\phi = 0$ ,  $\chi = 1$ , and keeps the better candidate. As published, it evaluates both candidates of each follower and then re-evaluates the kept ones with the new population, so an iteration costs  $2N_p$  evaluations and LX-SSA runs half as many iterations as SSA; comparing the two tests LX-SSA as published, not the Laplace step alone. Pseudocode: Section S-I-B.

### C. Basic Variable Neighborhood Search

The basic VNS [40], [41], in the continuous form of [49], improves one incumbent  $\mathbf{X}$  with  $k_{\max} = 5$  nested  $\ell_\infty$  neighborhoods  $\mathcal{N}_k(\mathbf{X}) = \{\mathbf{X}' : \delta_{k-1} < \|\mathbf{X}' - \mathbf{X}\|_\infty \leq \delta_k\}$ ,  $\delta_k = 0.1 k r$ , of the whole layout. Shaking draws a uniform point of  $\mathcal{N}_k(\mathbf{X})$ , clipped to the box, and a best-improvement local search improves it. As the top-hat wake makes  $F_p$  discontinuous, this is a compass search: it evaluates all  $2n$  moves  $\pm h \mathbf{e}_m$  ( $\mathbf{e}_m$  the  $m$ th unit vector), moves to the best improving one, and halves  $h$  (from  $0.05r$ ) when none improves, until  $h < 10^{-3}r$ .

## IV. TWO-PHASE SWARM-VNS DESIGN AND PSO-VNS

### A. Two-Phase Template

All hybrids follow one sequential (relay) template. For a budget of  $B$  evaluations and a split  $\omega \in (0, 1)$ , Phase 1 starts from the seeded initial population, runs a population method for about  $\omega B$  evaluations and passes its best layout to Phase 2, the basic VNS, which applies the local search to it and then iterates until  $B$  evaluations are used. Only Phase 1 differs: PSO in *PSO-VNS*; SSA or LX-SSA in *SSA-VNS* and *LX-SSA-VNS* (with  $\xi_1$  compressed to the Phase-1 iterations); and, in the controls,  $\text{round}(\omega B)$  layouts: the initial population, then layouts drawn uniformly in the farm disc (*RSD-VNS*) or in its bounding square (*RS-VNS*). VNS alone is the limiting case in which Phase 1 is the initial population. PSO and SSA run for  $T_1 = \text{round}((\omega B - N_p)/N_p)$  Phase-1 iterations, LX-SSA for  $\text{round}((\omega B - N_p)/(2N_p))$ . A swarm phase is useful only if it gives VNS a better start than random sampling with about the same number of evaluations (at  $\omega = 0.5$ , 3,015 for the controls and  $N_p(T_1 + 1) = 3,030$  for the swarms); the component analysis (Section VII) tests this for each Phase-1 choice against *RSD-VNS*, with *RS-VNS* as a weak minimum bar: a square sample is feasible with probability at most  $(\pi/4)^N$  and, mainly because of the spacing, below  $10^{-6}$  for the largest cases (Table S51); 730 *RS-VNS* runs have no feasible sample, and sampling in the disc gives one in only 242 of them.

### B. PSO-VNS

PSO-VNS (Algorithm 1) is not a new PSO variant: its Phase 1 is the plain PSO of Section III-A and, as the coefficients do not depend on the iteration, reproduces exactly, for a given seed, the first  $N_p(T_1 + 1)$  evaluations of a PSO run; comparing PSO-VNS with PSO thus isolates the effect of Phase 2.

PSO-VNS uses  $N_p = 30$ , the coefficients (7) and the VNS settings above, and adds one parameter, the split  $\omega$ . With the default  $\omega = 0.5$  and  $B = 6,030$ , PSO runs for  $T_1 = 100$  iterations (99.5 rounded half to even) and VNS receives the

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### Algorithm 1 PSO-VNS for a prescribed turbine count $N$

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**Require:** budget  $B$ , split  $\omega$ ,  $N_p$ ,  $w$ ,  $c_1$ ,  $c_2$ , radius  $r$ ,  $k_{\max}$

- 1: Draw and evaluate  $N_p$  particles  $\mathbf{X}^i$ ;  $\mathbf{V}^i \leftarrow \mathbf{0}$ ;  $\mathbf{P}^i \leftarrow \mathbf{X}^i$ ;  
 $\mathbf{G} \leftarrow \arg \min_i F_p(\mathbf{P}^i)$  ▷ Phase 1
- 2: **for**  $t = 1, \dots, T_1$  and  $i = 1, \dots, N_p$  **do**
- 3:   Move  $\mathbf{X}^i$  by (5), (6); box repair; evaluate  $F_p(\mathbf{X}^i)$ ;  
     update  $\mathbf{P}^i$ ,  $\mathbf{G}$
- 4: **end for**
- 5:  $\mathbf{X} \leftarrow \text{LOCALSEARCH}(\mathbf{G})$ ;  $k \leftarrow 1$  ▷ Phase 2
- 6: **while** fewer than  $B$  evaluations have been used **do**
- 7:    $\mathbf{X}' \leftarrow \text{LOCALSEARCH}(\text{uniform point of } \mathcal{N}_k(\mathbf{X}))$
- 8:   **if**  $F_p(\mathbf{X}') < F_p(\mathbf{X})$  **then**  $\mathbf{X} \leftarrow \mathbf{X}'$ ,  $k \leftarrow 1$  **else**  $k \leftarrow$   
      $k \bmod k_{\max} + 1$
- 9: **end while**
- 10: **return** best layout evaluated

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remaining 3,000 evaluations. None of these values was tuned; Section VII studies  $\omega$ . At this budget, PSO-VNS is essentially PSO plus one compass-search descent: the first descent is unfinished in 39 of 40 runs with  $N \geq 12$ , and shaking-plus-search cycles take 21.5% of the Phase-2 evaluations (pooled; median run 4.7%) for 0.5% of the Phase-2 gain (accepted cycles), so the neighborhoods matter only for small  $N$ .

## V. EVALUATION PROTOCOL

### A. Cases and Methods

The 68 cases combine the two Kusiak–Song wind data sets [5], [8] (Table S1) with circular farms of radii  $r = 500$ , 750 and 1000 m and  $N = 2, \dots, 10$ ,  $2, \dots, 12$  and  $2, \dots, 15$  turbines ( $R = 38.5$  m,  $\ell_{\min} = 4D = 308$  m,  $C_T = 0.8$ ,  $K = 0.075$ ).

The *main comparison* includes eight methods ( $N_p = 30$  where applicable): PSO-VNS (Algorithm 1,  $\omega = 0.5$ ); PSO; SSA-VNS; SSA; LX-SSA; DE/rand/1/bin [50] (scale factor 0.5, crossover rate 0.9); VNS; and MS-SLSQP, a multi-start sequential least-squares programming method (SciPy's SLSQP [51], forward-difference gradients) that minimizes the wake loss under explicit constraints, restarting from the initial, then random layouts (Section S-I); all others minimize  $F_p$  (4) over  $[-r, r]^{2N}$ . The *component analysis* (Section VII) adds LX-SSA-VNS and two random-sampling controls (Section IV-A; *RSD-VNS* on the benchmark only): after the common initial population, *RSD-VNS* samples Phase 1 uniformly in the farm disc and *RS-VNS* in the bounding square, 3,015 layouts versus the hybrids' 3,030 Phase-1 evaluations. A split study varies  $\omega$  on twelve cases (Section VII).

### B. Comparison Design and Statistics

*Equal budgets and paired seeds.* Unless stated otherwise, a run uses exactly  $B = 6,030$  objective evaluations, including the initial population and MS-SLSQP's gradient evaluations. Each method uses seeds 1–30 and draws its initial population first, so runs with equal seeds share it and are paired.

*Feasibility-aware ranking.* Within a case, a method is ranked by the mean benchmark objective of its feasible runs only if at least 15 of its 30 runs (5 of 10 for 10 seeds) are feasible, and

otherwise below all such methods, by its number of feasible runs. This limits, but does not remove, survivorship bias.

*Statistics.* All tests are two-sided ( $\alpha = 0.05$ ), with Holm's correction [52] within each family of comparisons. 1) Case ranks: the Friedman test with the Iman–Davenport F test and post hoc tests of the average ranks against PSO-VNS [21], [22], [24]. 2) Case means (primary, as mean-rank post hoc tests depend on the pool [39]): Wilcoxon signed-rank tests of PSO-VNS against each method on the per-case mean wake losses, with Hodges–Lehmann estimates for key pairs (Table S30); a case in which only one of the two methods has at least 15 feasible runs counts as a maximal difference in its favor (cases: 2 for SSA, 2 for LX-SSA and 18 for DE), and one in which neither has is dropped (0 cases). Mean differences (both qualified) carry 95% bootstrap CIs. 3) Runs: per case, Holm-corrected Wilcoxon signed-rank tests on the 30 seed-paired runs (infeasible runs last, ordered by minimum spacing), with the rank-biserial correlation  $r_{rb}$  and counts of significant wins, ties and losses (W/T/L). 4) Equivalence [26]: TOST (90% interval of the mean case-mean difference within  $\pm m$ ) at three levels (Table S26): *seed level*, cases fixed and runs resampled within them (estimand: the benchmark-average difference); *case level*, cases resampled (the mean over similar cases); *cluster level*, new farm clusters (CR2 interval, 5 d.f.; wild-cluster bootstrap- $t$ , Webb weights). The margin  $m = 0.05$  pp (4.1 and 2.1 MWh/yr per turbine in Data Sets I and II) is a post hoc convention; the minimal margins  $m_{\min}$  let readers apply their own. A Bayesian signed-rank test with this region [25] gives the posterior probability that equivalence is the most probable of three outcomes (Table S15).

*Robustness of inference.* The 68 cases form six (data set, radius) clusters with nested  $N$ . Six qualification thresholds (1, 10, 15, 20, 25 and 30 feasible runs) change no verdict; cluster analyses (sign per cluster, cluster-robust  $t$ , cluster bootstrap, leave-one-out) qualify only the significant contrasts of the salp-swarm hybrids with the controls; one Holm family over all 28 case-mean tests makes only PSO-VNS vs. PSO,  $N \geq 10$ , both data sets and PSO-VNS ( $\omega = 0.5$ ) vs. PSO-VNS ( $\omega = 0.9$ ) nonsignificant (Tables S17–S22); Section IX adds a  $1^\circ$  direction resolution.

### C. Tests Beyond the Benchmark and Reproducibility

*Initialization and budget.* The feasible initialization draws seeded layouts inside the site and separates the turbines with L-BFGS-B on a packing penalty, without objective evaluations (Section S-I); all methods but RS-VNS use it at 6,030 evaluations. With random starts, all ten run at 6,030, 30,030 and 120,030 evaluations. Both studies use the six largest cases (largest  $N$  per farm and data set) and the Horns Rev 1 block, with 30 seeds (10 for Horns Rev 1 at 120,030).

*Other sites and models.* The Horns Rev 1 block is PyWake's 16-turbine example [53]: V80 turbines ( $D = 80$  m), the measured 12-sector wind climate, the parallelogram through the corner turbines as boundary (Section VIII-B),  $\ell_{\min} = 4D$  and the Jensen wake with  $K = 0.04$  and  $C_T$  at the free-stream speed. The 16- and 36-turbine scenarios of IEA37 Case Study 1 [32], [54] (circular sites, radii 1300 and 2000 m,

TABLE I  
EFFECT OF THE PSO SETTING ON AN EIGHT-METHOD POOL (THE METHODS OF TABLE II WITH LX-SSA-VNS INSTEAD OF PSO-VNS; 68 CASES, 6,030 EVALUATIONS)

| Method                     | PSO setting |              |
|----------------------------|-------------|--------------|
|                            | Old         | Constriction |
| LX-SSA-VNS                 | 2.65        | 3.41         |
| SSA-VNS                    | <b>1.99</b> | 2.77         |
| LX-SSA                     | 5.65        | 6.06         |
| SSA                        | 4.81        | 5.34         |
| PSO                        | 5.04        | <b>1.53</b>  |
| DE                         | 6.99        | 7.03         |
| VNS                        | 3.59        | 4.15         |
| MS-SLSQP                   | 5.28        | 5.71         |
| PSO position               | 5           | 1            |
| LX-SSA-VNS vs. PSO (W/T/L) | 35/33/0     | 0/21/47      |
| SSA-VNS vs. PSO (W/T/L)    | 41/27/0     | 0/22/46      |

Old:  $w = 0.7$ ,  $c_1 = c_2 = 2$ ; constriction: Clerc–Kennedy setting [46]. Average rank: 1 = best, feasibility-aware rule (fewer than 15 of 30 feasible runs in a case = ranked last); bold: best average rank. W/T/L: cases in which the hybrid is significantly better / not different / worse than PSO (two-sided Wilcoxon signed-rank test, 30 seed-paired runs, Holm-adjusted over the seven comparisons of the hybrid in each case,  $\alpha = 0.05$ ).

$\ell_{\min} = 2D$ , 16 directions at one speed, a simplified Bastankhah Gaussian wake) test optimizers rather than represent a site; the eight methods of the main comparison run on them at 6,030 and 30,030 evaluations (30 seeds).

*Verification and code.* Our objective matches the original implementation (largest difference  $8.7 \times 10^{-11}$  over 720 archived final objectives) and the official IEA37 calculator (relative difference  $< 10^{-11}$ ); our reused SSA, LX-SSA, PSO and DE code reproduces the recorded result distributions of the original code (2 of 24 Mann–Whitney tests reject at  $\alpha = 0.05$ ; 1.2 expected), and instrumented copies reproduce the stored runs (522 of 522). Runs: Linux containers (Intel Xeon; Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1); code, per-run results and scripts are public (Data Availability).

## VI. RESULTS ON THE BENCHMARK

This section reports the 68 benchmark cases (22,440 runs of 6,030 evaluations, random starts,  $15^\circ$  rose; per case: Tables S4–S9).

### A. Effect of the Baseline Setting

The old setting lies beyond the order-2 (variance) stability boundary (8). Run on the same cases, budget and seeds as the constriction setting (7), it returns a feasible layout in only 86.0% of the runs (fewer than 15 feasible runs in 8 cases), is significantly worse in 56 of the 68 cases and never better, and has a mean wake loss 0.61 pp higher over the 60 cases in which both settings have at least 15 feasible runs. In instrumented reruns (clipping without velocity reset or clamping; Table S52), its swarm contracts much less (final spread 8.9 times, velocity 13 times that of the constriction setting), 8.7% of the coordinates are clipped (constriction: 1.6%), and far fewer particle evaluations in the second half of a run are feasible (2.6% vs. 48.4%). A sweep of  $c_1 = c_2 = c$  at  $w = 0.7$  (Fig. S1) shows that the boundary  $c^* = 1.7486$  marks

TABLE II  
CASE-LEVEL ANALYSIS OF THE EIGHT METHODS OVER THE 68  
BENCHMARK CASES

| Method   | Avg. rank | Best | Feas. | $p_z$                 | $p_W$                 | $\Delta\bar{L}$ |
|----------|-----------|------|-------|-----------------------|-----------------------|-----------------|
| PSO-VNS  | 1.82      | 29   | 100.0 | —                     | —                     | —               |
| PSO      | 1.97      | 25   | 100.0 | 0.713                 | 0.296                 | -0.018          |
| SSA-VNS  | 3.45      | 0    | 99.6  | $2.0 \times 10^{-4}$  | $5.7 \times 10^{-11}$ | -0.330          |
| VNS      | 4.40      | 3    | 100.0 | $2.4 \times 10^{-9}$  | $1.3 \times 10^{-10}$ | -0.465          |
| SSA      | 5.38      | 0    | 97.6  | $8.3 \times 10^{-17}$ | $4.5 \times 10^{-11}$ | -0.611          |
| MS-SLSQP | 5.88      | 1    | 100.0 | $1.8 \times 10^{-21}$ | $4.2 \times 10^{-11}$ | -0.435          |
| LX-SSA   | 6.06      | 0    | 96.9  | $3.3 \times 10^{-23}$ | $4.5 \times 10^{-11}$ | -0.765          |
| DE       | 7.04      | 0    | 71.4  | $1.0 \times 10^{-34}$ | $5.6 \times 10^{-11}$ | -0.887          |

Avg. rank: average rank of the mean feasible objective (1 = best; fewer than 15 feasible runs in a case = ranked last); Best: cases in which the method alone ranks first; Feas.: feasible runs (%);  $p_z$ : Holm-adjusted  $p$  of the average-rank test against PSO-VNS. Because mean-rank tests depend on the pool [39],  $p_W$  gives the two-sided Wilcoxon signed-rank test on the 68 per-case mean wake losses (Holm-adjusted over the 7 methods) and  $\Delta\bar{L}$  the mean wake-loss difference (pp, PSO-VNS minus method; negative = PSO-VNS better) over the cases in which both methods have at least 15 feasible runs (66 for SSA, 66 for LX-SSA, 50 for DE, otherwise 68). Friedman  $\chi^2_F = 319.3$  (7 d.f.),  $p = 4.6 \times 10^{-65}$ ; Iman-Davenport  $F_F = 136.5$  (7, 469 d.f.),  $p = 6.7 \times 10^{-109}$ .

TABLE III  
RUN-LEVEL OUTCOME OF PSO-VNS AGAINST EACH METHOD (W/T/L  
OVER THE 68 CASES)

| DS             | $r$ (m) | Cases | PSO     | SSA-VNS | SSA    | LX-SSA | DE      | VNS     | MS-SLSQP |
|----------------|---------|-------|---------|---------|--------|--------|---------|---------|----------|
| I              | 500     | 9     | 0/8/1   | 6/3/0   | 8/1/0  | 8/1/0  | 7/2/0   | 6/3/0   | 7/2/0    |
| I              | 750     | 11    | 1/8/2   | 9/2/0   | 10/1/0 | 10/1/0 | 9/2/0   | 8/3/0   | 10/1/0   |
| I              | 1000    | 14    | 3/11/0  | 9/5/0   | 12/2/0 | 12/2/0 | 12/2/0  | 12/2/0  | 12/2/0   |
| II             | 500     | 9     | 3/5/1   | 7/2/0   | 7/2/0  | 8/1/0  | 7/2/0   | 7/1/1   | 8/1/0    |
| II             | 750     | 11    | 2/9/0   | 8/3/0   | 10/1/0 | 10/1/0 | 9/2/0   | 8/3/0   | 9/2/0    |
| II             | 1000    | 14    | 5/9/0   | 10/4/0  | 12/2/0 | 12/2/0 | 12/2/0  | 10/4/0  | 13/1/0   |
| All            |         | 68    | 14/50/4 | 49/19/0 | 59/9/0 | 60/8/0 | 56/12/0 | 51/16/1 | 59/9/0   |
| $\bar{r}_{rb}$ |         |       | +0.08   | +0.72   | +0.98  | +0.97  | +1.00   | +0.84   | +0.99    |

W/T/L: cases in which PSO-VNS is significantly better / not different / worse (two-sided Wilcoxon signed-rank test on 30 seed-paired runs,  $\alpha = 0.05$ , Holm-adjusted over the seven comparisons of PSO-VNS in each case, a different family from Table IV; an infeasible run ranks below every feasible run).  $\bar{r}_{rb}$ : median rank-biserial correlation (positive = PSO-VNS better; zero differences dropped,  $\bar{r}_{rb} = 0$  if all are zero).

approximately where this PSO starts to fail: at least 98.9% of the runs are feasible for  $c \leq 1.7$ , then feasibility declines progressively (90.0% at  $c = 1.8$ , 58.9% at 2.0); velocity clamping raises it to 92.2%, still worse than constriction in all 12 cases.

Table I shows the effect on a comparison. It ranks an eight-method pool, the methods of Table II with LX-SSA-VNS instead of PSO-VNS, once with each PSO setting (all other methods use identical runs): with the old setting, SSA-VNS ranks first and PSO only in position 5, behind both salp-swarm hybrids; with the constriction setting, PSO ranks first, ahead of every salp-swarm method (standard coefficients [46]; none of the methods was tuned on these cases). The reversal holds at 6,030 evaluations; on the six largest cases at 120,030, SSA-VNS ranks ahead of PSO (4.17 vs. 5.83; Section VIII-A). A PSO baseline beyond the order-2 boundary thus makes methods compared with it, including our own LX-SSA, appear stronger than they are.

### B. Overall Comparison

Over the eight methods (Table II), the Friedman test rejects the hypothesis of equal performance ( $\chi^2_F = 319.3$ ,  $p = 4.6 \times 10^{-65}$ ; Iman-Davenport  $F_F = 136.5$ ,  $p = 6.7 \times 10^{-109}$ ). PSO-

VNS has the best average rank (1.82), alone ranks first in 29 cases, and stays first when the final layouts are re-evaluated with  $5^\circ$  or  $1^\circ$  direction sub-bins ( $1^\circ$ : 2.09; Section IX). In the Holm-adjusted post hoc tests ( $p_z$  in Table II; tied ranks make them conservative), PSO-VNS ranks significantly better than SSA-VNS, VNS, SSA, MS-SLSQP, LX-SSA and DE, but not than PSO ( $p_{\text{Holm}} = 0.71$ ). The rank order and these conclusions are unchanged for thresholds of 1, 10, 15, 20, 25 and 30 feasible runs; significant comparisons with PSO-VNS favor the same method in each of the six clusters, and leaving out one cluster changes none of them (Tables S17–S20).

Against PSO, PSO-VNS is significantly better in 14 cases, tied in 50 and worse in 4 (run level, Table III). The case means do not differ significantly ( $\Delta\bar{L} = -0.018$  pp,  $p = 0.30$ ; Hodges-Lehmann estimate  $-0.003$  pp). Whether the difference is negligible at the post hoc margin  $\pm 0.05$  pp depends on the level of inference (Table S26): on this fixed benchmark (seed level), PSO-VNS has a small significant advantage inside the margin (90% CI  $[-0.026, -0.010]$ ); over exchangeable cases the two are equivalent ( $[-0.041, 0.004]$ , TOST  $p = 0.011$ , minimal margin 0.041); over farm clusters, equivalence is not shown (CR2  $[-0.067, 0.031]$ , minimal margin 0.067). In the Bayesian signed-rank test, practical equivalence is the most probable outcome (posterior probability  $> 0.99$ ; posterior means 0.25/0.62/0.12 for PSO-VNS better by more than the margin/equivalent/PSO better). The margin corresponds to 4.1 (Data Set I) and 2.1 MWh/yr (Data Set II) per turbine, the mean gain of PSO-VNS to 0.2 MWh/yr per turbine (0.020% of AEP). Spread and worst run do not differ ( $p = 0.55, 0.80$ ).

The average hides case-level differences in both directions: 11 cases favor PSO-VNS by more than the margin (by up to 0.41 pp) and 8 favor PSO (up to 0.27 pp; Table S28). The two are equivalent for  $N < 10$  (48 cases, 90% CI  $[-0.005, 0.021]$ ) but not for  $N \geq 10$  ( $-0.079$  pp,  $[-0.142, -0.017]$ ) or on the 44 cases with a wake loss of at least 0.2 pp ( $[-0.060, 0.008]$ ). They follow a data set  $\times$  density crossover (density  $\phi = N(\ell_{\min}/2r)^2$ ; stated after review, exploratory; interaction  $p = 0.01$ ): as  $\phi$  grows, PSO-VNS gains in Data Set II (slope  $-0.52$  pp,  $p = 1.0 \times 10^{-4}$ ) and PSO in Data Set I ( $+0.29$  pp,  $p = 0.0012$ ). In the post hoc subgroup Data Set II,  $N \geq 10$ , PSO-VNS has the lower mean wake loss in 10 of 10 cases ( $p_{\text{Holm}} \leq 0.018$  over all 28 case-mean Wilcoxon tests; 0.221% of AEP); all 7 Data Set I cases beyond the margin favor PSO. With  $1^\circ$  sub-bins, PSO-VNS is slightly but significantly better than PSO ( $\Delta\bar{L} = -0.043$  pp, 90% CI  $[-0.062, -0.025]$ ; Table S43).

Against SSA-VNS, PSO-VNS is significantly better in 49 cases and never worse ( $p_{\text{Holm}} = 5.7 \times 10^{-11}$ ). Against SSA, LX-SSA, DE, VNS and MS-SLSQP (untuned DE; finite-difference gradients), it is significantly better in most cases; it is never significantly worse than SSA-VNS, SSA, LX-SSA, DE and MS-SLSQP and significantly worse than VNS in one case. PSO-VNS is more reliable in reaching feasibility on the Horns Rev 1 block with random starts (30/30 vs. 19/30, McNemar  $p = 9.8 \times 10^{-4}$ ; Section VIII-B) and ranks best on the six largest cases at all three budgets (exploratory; Section VIII-A). The leading methods depend on the conditions: from feasible starts, VNS ranks first (Section VIII-A); under the Gaussian

TABLE IV  
COMPONENT ANALYSIS OVER THE 68 BENCHMARK CASES (6,030  
EVALUATIONS, 30 SEED-PAIRED RUNS)

| Variant    | Phase 1         | Phase 2 | Avg. rank | Feas. (%) |
|------------|-----------------|---------|-----------|-----------|
| PSO-VNS    | PSO             | VNS     | 1.96      | 100.0     |
| SSA-VNS    | SSA             | VNS     | 4.46      | 99.6      |
| LX-SSA-VNS | LX-SSA          | VNS     | 5.41      | 99.9      |
| RS-VNS     | random (square) | VNS     | 5.29      | 99.4      |
| RSD-VNS    | random (disc)   | VNS     | 3.77      | 99.8      |
| VNS        | —               | VNS     | 6.12      | 100.0     |
| PSO        | PSO             | —       | 2.24      | 100.0     |
| SSA        | SSA             | —       | 7.50      | 97.6      |
| LX-SSA     | LX-SSA          | —       | 8.25      | 96.9      |

| Contrast               | Isolates                           | W/T/L   | $p_W$                 | $\overline{\Delta L}$ |
|------------------------|------------------------------------|---------|-----------------------|-----------------------|
| PSO-VNS vs. PSO        | VNS phase (PSO)                    | 8/58/2  | 0.592                 | −0.018                |
| SSA-VNS vs. SSA        | VNS phase (SSA)                    | 39/29/0 | $5.7 \times 10^{-10}$ | −0.294                |
| LX-SSA-VNS vs. LX-SSA  | VNS phase (LX-SSA)                 | 34/34/0 | $1.1 \times 10^{-10}$ | −0.394                |
| PSO-VNS vs. VNS        | swarm start vs. best initial point | 49/18/1 | $5.7 \times 10^{-10}$ | −0.465                |
| PSO-VNS vs. RS-VNS     | PSO vs. random sampling            | 47/21/0 | $3.2 \times 10^{-10}$ | −0.399                |
| SSA-VNS vs. RS-VNS     | SSA vs. random sampling            | 2/66/0  | 0.002                 | −0.068                |
| LX-SSA-VNS vs. RS-VNS  | LX-SSA vs. random sampling         | 1/67/0  | 0.718                 | −0.005                |
| PSO-VNS vs. RSD-VNS    | PSO vs. disc sampling              | 45/23/0 | $4.0 \times 10^{-10}$ | −0.306                |
| SSA-VNS vs. RSD-VNS    | SSA vs. disc sampling              | 0/66/2  | 0.014                 | +0.024                |
| LX-SSA-VNS vs. RSD-VNS | LX-SSA vs. disc sampling           | 1/62/5  | $8.2 \times 10^{-6}$  | +0.087                |
| RSD-VNS vs. RS-VNS     | disc vs. square sampling           | 6/62/0  | $4.2 \times 10^{-7}$  | −0.093                |
| PSO-VNS vs. SSA-VNS    | PSO vs. SSA as Phase 1             | 44/24/0 | $2.5 \times 10^{-10}$ | −0.330                |
| PSO-VNS vs. LX-SSA-VNS | PSO vs. LX-SSA as Phase 1          | 47/21/0 | $2.3 \times 10^{-10}$ | −0.393                |
| SSA-VNS vs. LX-SSA-VNS | LX-SSA as published (hybrid)       | 2/66/0  | $2.7 \times 10^{-5}$  | −0.063                |
| LX-SSA vs. SSA         | LX-SSA as published (alone)        | 0/64/4  | $1.3 \times 10^{-6}$  | +0.153                |

Top: average rank among the nine variants (Friedman  $\chi^2_F = 374.3$ , 8 d.f.,  $p = 5.96 \times 10^{-76}$ ) and feasible runs. Bottom: planned contrasts. W/T/L: cases in which the first variant is significantly better / not different / worse (run-level Wilcoxon,  $\alpha = 0.05$ , Holm-adjusted over the fifteen contrasts of each case, a different family from Table III);  $p_W$ : Wilcoxon test on the 68 per-case mean wake losses, Holm-adjusted over the fifteen contrasts;  $\overline{\Delta L}$ : mean difference of the case-mean wake losses (pp; negative = first variant better) over the cases in which both variants have at least 15 feasible runs (66–68 of 68 per contrast).

wake, PSO ranks narrowly ahead of PSO-VNS with  $15^\circ$  bins but not with  $1^\circ$  sub-bins (Section IX).

The wake loss grows with  $N$ , falls as the farm becomes larger, and is higher for the broader wind rose of Data Set II (Section S-II-E). PSO-VNS ends feasible in all runs (DE: 71.4%). In the densest cases (500 m,  $N = 10$ ), the global best of the PSO phase is feasible at the switch in only 63.3% of the runs, but all runs are feasible after the VNS phase.

After the switch, both keep improving (Fig. 1): the VNS phase removes on average 30.3% (median 22.6%) of the wake loss left after the PSO phase and repairs 26 runs infeasible at the switch, whereas continuing PSO for the same 3,000 evaluations removes 29.2% (median 20.6%).

## VII. COMPONENT ANALYSIS OF TWO-PHASE HYBRIDS

The variants of Table IV combine a Phase 1 (PSO, SSA, LX-SSA, random sampling or none) with or without the same VNS phase (6,030 evaluations from random starts, same seeds;

convergence: Fig. 1); their average ranks differ (Friedman  $p = 6.0 \times 10^{-76}$ ). W/T/L,  $p$  (Holm-adjusted Wilcoxon test on the 68 case means) and  $\overline{\Delta L}$  (mean difference of the case means, pp) are those of Table IV; a 90% bootstrap interval of  $\overline{\Delta L}$  (in brackets) inside  $\pm 0.05$  pp means practical equivalence.

*VNS phase.* The VNS phase improves both salp-swarm methods: SSA-VNS is 39/29/0 (W/T/L) against SSA and LX-SSA-VNS 34/34/0 against LX-SSA ( $\overline{\Delta L} = -0.294$  and  $-0.394$ ). For PSO, which PSO-VNS repeats for its first 3,030 evaluations (Section IV), the difference is small, not significant and practically equivalent to zero ( $\overline{\Delta L} = -0.018$  [−0.041, 0.004],  $p = 0.59$ ). At this budget the VNS phase is essentially one compass-search descent: for PSO-VNS, the shake-and-search cycles take 21.5% of the Phase-2 evaluations (pooled; median run 4.7%) but give 0.5% of the Phase-2 gain (2.5% with the cycle cut off by the budget), and the median run with  $N \geq 12$  never shakes (Table S53). The analysis thus measures the value of a local-search phase.

*Controls.* Random sampling of equal cost replaces Phase 1, uniform in the site disc (RSD-VNS) or in its bounding square (RS-VNS); VNS alone is a third control. RS-VNS is weak: 730 of its runs draw no feasible sample, although all of them draw samples wholly inside the circle: the spacing constraint is binding, and disc sampling rescues only 242 of these runs. RSD-VNS beats RS-VNS (6/62/0,  $\overline{\Delta L} = -0.093$ ) and ranks third (3.77).

*Swarm phase.* A PSO phase clearly beats every control: PSO-VNS is 45/23/0 against RSD-VNS ( $\overline{\Delta L} = -0.306$  [−0.380, −0.239], wholly below −0.05), 47/21/0 against RS-VNS and 49/18/1 against VNS. At this budget and from random starts, salp-swarm phases add nothing beyond random sampling in the farm. Against RSD-VNS, SSA-VNS is 0/66/2 ( $\overline{\Delta L} = +0.024$ ; Hodges–Lehmann estimate +0.025 pp, 95% CI [0.007, 0.058]; the case means, not the farm clusters, favor RSD-VNS): a gain of the SSA phase larger than the margin is ruled out at the seed and the case level (90% intervals [0.002, 0.046] and [−0.022, 0.062], both above −0.05), whereas equivalence remains inconclusive. LX-SSA-VNS is 1/62/5 and worse ( $\overline{\Delta L} = +0.087$ ,  $p = 8.2 \times 10^{-6}$ ). The small SSA gain over RS-VNS (2/66/0,  $\overline{\Delta L} = -0.068$ ) only reflects that control’s weakness; LX-SSA-VNS and RS-VNS are practically equivalent (−0.034, 0.021). Re-evaluated with  $1^\circ$  direction bins (Table S44), SSA-VNS and LX-SSA-VNS again have a higher loss than RSD-VNS (0.074 and 0.135 pp) and PSO-VNS beats all three controls; only the SSA gain over RS-VNS is no longer significant.

*LX-SSA as published.* LX-SSA costs  $2N_p$  evaluations per iteration (Section III-B) and so runs half as many iterations as SSA; the contrasts do not separate its Laplace step from this cost. As published, it makes the salp swarm worse: LX-SSA is 0/64/4 against SSA ( $\overline{\Delta L} = +0.153$ ,  $p = 1.3 \times 10^{-6}$ ) and SSA-VNS is 2/66/0 against LX-SSA-VNS ( $\overline{\Delta L} = -0.063$ ,  $p = 2.7 \times 10^{-5}$ ).

*Choice of Phase 1.* With the same VNS phase, PSO-VNS is 44/24/0 against SSA-VNS and 47/21/0 against LX-SSA-VNS. At the switch, its global best is feasible more often (98.7% of the runs vs. 95.7% and 94.1%) and, when feasible, has a lower wake loss (1.63% vs. 2.01% and 2.11%). Whether PSO helps

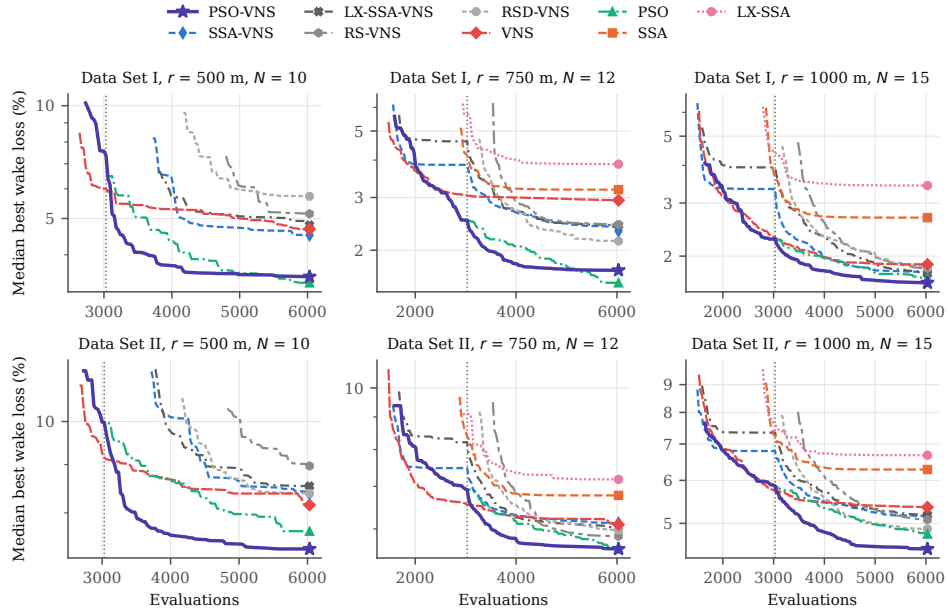


Fig. 1. Convergence of the nine component-analysis variants (Table IV) for the largest turbine count of each farm: median best feasible wake loss over 30 runs versus evaluations. PSO-VNS and PSO coincide up to 3,030 evaluations (dotted line), where PSO-VNS switches to VNS. Curves of the main comparison: Fig. S4 of the supplementary material.

TABLE V  
INITIALIZATION AND BUDGET ON THE SIX LARGEST BENCHMARK CASES  
(30 SEEDS)

| Method     | Random init. |       | Feasible init. |       | Avg. rank |        |         |
|------------|--------------|-------|----------------|-------|-----------|--------|---------|
|            | Loss         | Feas. | Loss           | Feas. | 6,030     | 30,030 | 120,030 |
| PSO-VNS    | 4.084        | 100.0 | 3.938          | 100.0 | 1.50      | 1.17   | 2.17    |
| SSA-VNS    | 4.678        | 95.0  | 3.955          | 100.0 | 4.33      | 4.17   | 4.17    |
| LX-SSA-VNS | 4.802        | 98.9  | 3.952          | 100.0 | 5.17      | 5.67   | 5.83    |
| RS-VNS     | 4.830        | 93.3  | n/r            | n/r   | 4.67      | 6.33   | 4.67    |
| VNS        | 4.950        | 100.0 | 3.810          | 100.0 | 6.50      | 4.83   | 3.17    |
| PSO        | 4.181        | 100.0 | 5.549          | 100.0 | 1.83      | 2.50   | 5.83    |
| SSA        | 4.828        | 80.0  | 4.947          | 100.0 | 8.33      | 8.67   | 8.83    |
| LX-SSA     | 5.379        | 77.8  | 5.048          | 100.0 | 8.67      | 9.83   | 10.00   |
| DE         | —            | 1.1   | 5.549          | 100.0 | 10.00     | 5.17   | 3.33    |
| MS-SLSQP   | 4.522        | 99.4  | 4.294          | 100.0 | 4.00      | 6.67   | 7.00    |

Loss: mean wake loss (%) of the feasible runs; Feas.: feasible runs (%); both at 6,030 evaluations with random and feasibility-preserving initialization. Avg. rank at 6,030, 30,030 and 120,030 evaluations, random initialization. n/r: not run.

mainly by faster feasibility or by better exploration remains open (cf. the feasible starts in Section VIII-A).

**Budget split.** Of five splits tested on twelve cases (from  $\omega = 0.25$  to plain PSO,  $\omega = 1$ ; Table S12),  $\omega = 0.75$  has the lowest mean wake loss and average rank and  $\omega = 0.9$  does not beat it, so a split above the preset  $\omega = 0.5$  used here is a candidate to confirm on the target problem.

## VIII. TESTS BEYOND THE BENCHMARK

### A. Budget Scaling and Feasibility-Preserving Initialization

**Budget.** On the six largest benchmark cases (random starts), PSO-VNS keeps the best average rank at all three budgets (6,030, 30,030 and 120,030 evaluations: 1.50, 1.17 and 2.17; Table V; Fig. S11) and ranks first in 4, 5 and 3 of the six cases (exploratory; rank differences not tested); PSO's mean wake loss minus that of PSO-VNS is +0.10, +0.18 and +0.27 pp

(Table S33). The order of the other methods depends on the budget: from 6,030 to 120,030 evaluations, PSO falls back from an average rank of 1.83 to 5.83, behind SSA-VNS (4.17), and VNS moves up from 6.50 to 3.17; at 120,030, VNS, SSA-VNS, RS-VNS, LX-SSA-VNS and DE come within 0.17 pp of the mean wake loss of PSO-VNS (3.14%). The ranking at 6,030 evaluations is thus that of a small budget.

**Feasible starts.** With feasibility-preserving initialization, the share of feasible runs rises for SSA-VNS (95.0% to 100.0%), SSA (80.0% to 100.0%), LX-SSA (77.8% to 100.0%) and DE (1.1% to 100.0%). VNS alone is then best in mean wake loss and average rank, and PSO-VNS ranks second (rank 2.83; loss 3.94%, 4.08% with random starts; Table S32). PSO and DE never improve on the best initial layout (210 runs each): their moves change all or most coordinates at once and, between distinct dense layouts, make turbines collide (median minimum spacing after the first PSO move 95 m, limit 308 m), so in our runs no trial is both feasible and better and the global best never moves (Section S-VI-C; Table S54). VNS, which moves one coordinate at a time, is not affected. The advantage of PSO-VNS thus depends on the initialization: from random starts, PSO may help mainly by reaching feasibility.

### B. Horns Rev 1 16-Turbine Block

The model (Section S-I-G) uses the Jensen wake with a hub-center in-wake test, the thrust coefficient  $C_T$  at the free-stream speed, and  $5^\circ$  direction and 1-m/s speed bins; the boundary violation in (4) is the distance (m) outside the parallelogram. For the installed 80-turbine farm at identical bins, PyWake 2.6.20's NOJ model [53] gives 661.39 GWh/yr and ours 666.75 GWh/yr (0.81% more; wake loss 10.39% vs 11.11%); with  $C_T$  at the local speed, our model reproduces

TABLE VI  
HORNS REV 1 16-TURBINE BLOCK (5° DIRECTION BINS): AEP,  
FEASIBILITY AND AEP LOSS

| Layout / method  | 6,030 evaluations, R |       | Loss (%)           |      |                    |      |
|------------------|----------------------|-------|--------------------|------|--------------------|------|
|                  | AEP                  | Feas. | 6k R               | 6k F | 30k                | 120k |
| Installed layout | 139.82               | –     | 6.04               | –    | –                  | –    |
| PSO-VNS          | 138.29               | 30/30 | 7.07               | 6.97 | 6.52               | 6.40 |
| SSA-VNS          | 137.59               | 30/30 | 7.54               | 6.98 | 7.19               | 6.47 |
| LX-SSA-VNS       | 137.41               | 30/30 | 7.66               | 7.02 | 7.07               | 6.44 |
| RS-VNS           | 137.58               | 28/30 | 7.54 <sup>28</sup> | n/r  | 7.04               | 6.25 |
| VNS              | 137.35               | 30/30 | 7.70               | 6.91 | 6.85               | 6.23 |
| PSO              | 137.79               | 19/30 | 7.40 <sup>19</sup> | 8.35 | 6.78 <sup>28</sup> | 6.63 |
| SSA              | 136.25               | 30/30 | 8.44               | 7.83 | 8.17               | 7.62 |
| LX-SSA           | 136.00               | 28/30 | 8.61 <sup>28</sup> | 7.89 | 8.50               | 8.28 |
| DE               | –                    | 0/30  | –                  | 8.35 | 7.40 <sup>8</sup>  | 6.12 |
| MS-SLSQP         | 136.31               | 30/30 | 8.40               | 8.00 | 7.82               | 7.31 |

Wake-free AEP 148.81 GWh/yr. AEP: mean (GWh/yr) over the feasible runs of 30 seeds at 6,030 evaluations, random initialization (R); Feas.: feasible runs. Loss: mean AEP loss (%) of the feasible runs at 6,030 evaluations with random (R) and feasibility-preserving (F) initialization and at 30,030 and 120,030 evaluations (R; 10 seeds at 120,030); superscript: feasible runs when not all are feasible; n/r: not run. Run-level tests: supplementary material.

PyWake’s hub-center NOJ exactly. The installed layout, a regular  $7D$  grid, reaches 139.82 GWh/yr (wake loss 6.04%; Table VI; layouts: Fig. S12). We also re-evaluated all final layouts with  $1^\circ$  bins and with PyWake’s NOJ. At 6,030 evaluations from random starts, the robust result is feasibility: PSO-VNS is feasible in 30/30 runs, PSO in only 19/30 (exact McNemar  $p = 9.8 \times 10^{-4}$ ; 30/30 with feasible starts). The AEP advantage of PSO-VNS over PSO on the 19 jointly feasible seeds (+0.54 GWh/yr,  $p = 0.016$ ) vanishes with  $1^\circ$  bins ( $-0.05$ ,  $p = 0.65$ ) and under PyWake. At 30,030 evaluations, PSO-VNS has the highest mean AEP under all three evaluations (5° bins: 139.11 GWh/yr) and beats PSO on the 28 jointly feasible seeds also with  $1^\circ$  bins (+0.28 GWh/yr,  $p = 6.5 \times 10^{-4}$ ) and under PyWake; at 120,030 evaluations (10 seeds), DE has the highest mean AEP. With  $1^\circ$  bins, 0 of the 901 feasible optimized layouts (all methods and budgets) exceed the installed block (16 with 5° bins), although the looser spacing  $4D$  (installed:  $7D$ ) enlarges their feasible set. The block is evaluated without the other 64 turbines, and cables, foundations, loads and navigation are not modeled: the comparison tests the optimizers, not the installed design.

### C. IEA37 Case Study 1

In IEA37 Case Study 1 the participants chose their own optimizers and budgets [32]. None of our methods reaches the best feasible published layouts (Table VII; layouts: Section S-IV-C). Table VII counts a published layout as feasible at a tolerance of 1 mm (ours:  $10^{-6}$  m); after radial projection onto the boundary, 5 more are feasible, and the best are then those of participant 12. Against these, the best PSO-VNS layout at 30,030 evaluations differs in AEP by  $-1.93\%$  (16 turbines) and  $-6.23\%$  (36 turbines; strict:  $-1.33\%$  and  $-4.20\%$ ), in wake loss by +1.73 and +5.21 pp, and would rank fourth among 12 and ninth among 10 feasible submissions. With 36 turbines, the best of our layouts comes from MS-SLSQP ( $-5.56\%$ ). As the participants’ budgets are not reported comparably, the size of the gap is indicative only. The 64-turbine

TABLE VII  
IEA37 CASE STUDY 1: AEP (MWH) OF THE BEST RUN OF EACH  
METHOD AND OF PUBLISHED LAYOUTS [32], [54]

| Method / source                        | 16 turbines ( $r = 1300$ m) |           | 36 turbines ( $r = 2000$ m) |           |
|----------------------------------------|-----------------------------|-----------|-----------------------------|-----------|
|                                        | 6,030                       | 30,030    | 6,030                       | 30,030    |
| Example layout                         | 366,941.6                   |           | 737,883.1                   |           |
| Best published, strict <sup>a</sup>    | 418,924.4                   |           | 863,676.3                   |           |
| Best published, projected <sup>b</sup> | 421,451.1                   |           | 882,382.8                   |           |
| PSO-VNS                                | 402,782.4                   | 413,331.9 | 787,708.9                   | 827,385.0 |
| PSO                                    | 403,753.1                   | 412,702.4 | 787,988.7                   | 830,256.4 |
| SSA-VNS                                | 403,251.3                   | 406,924.6 | 766,803.3                   | 819,930.5 |
| SSA                                    | 395,150.2                   | 401,459.3 | 750,707.4                   | 802,538.2 |
| LX-SSA                                 | 393,284.6                   | 398,770.0 | 746,301.3                   | 789,311.3 |
| DE                                     | 357,391.3                   | 397,402.7 | –                           | –         |
| VNS                                    | 400,169.3                   | 404,076.0 | 785,791.3                   | 824,230.2 |
| MS-SLSQP                               | 405,433.8                   | 410,349.5 | 748,571.1                   | 833,365.0 |

AEP of the best feasible run of each method (30 seeds; –: no feasible run). <sup>a</sup>Best published layout with every turbine within 1 mm of the boundary (participant 4 in both scenarios). <sup>b</sup>Best published layout after projecting the turbines outside the boundary radially onto it, spacing re-checked (participant 12 in both scenarios; up to 3.5 m and 4.9 mm outside before projection).

scenario was not run, and, like the published layouts, ours are optimized for 16 discrete directions and one wind speed.

### D. Computational Cost and Published Results

Per evaluation, the metaheuristics need 0.23–0.45 ms and MS-SLSQP 0.53 ms, about 2.3 times as long as PSO (indicative; Table S37); one 6,030-evaluation run of PSO-VNS takes 1.5–3.2 s, depending on  $N$ . Published Kusiak–Song results [5]–[8] are discussed in Section S-IX; they are not compared numerically because budgets, constraint handling and reporting differ.

## IX. SENSITIVITY TO THE MODELING ASSUMPTIONS

We re-evaluated every feasible final layout of the main comparison, without re-optimization, under a cubic power curve [55] (also with a 25 m/s cut-out), the uncalibrated Gaussian wake of [56] ( $K_G = 0.04$ ), and a finer direction resolution: each  $15^\circ$  bin split into 15 sub-bins of  $1^\circ$  with the bin’s Weibull parameters (Section S-V-A; Tables S39 and S42). The cubic curve lowers the expected power by 17.7% (Data Set I) and 36.9% (Data Set II), so our energy values are benchmark values, but leaves the ranking intact ( $\bar{\tau} = 0.95$ ; same best method in 94% of the cases; PSO-VNS first). The  $1^\circ$  rose raises the mean wake loss of every method, by 0.40 pp (DE) to 1.04 pp (PSO; PSO-VNS: 1.322% to 2.335%); optimized layouts exploit the gaps between the coarse bins, those of PSO and the VNS-based methods most. PSO-VNS still ranks first (2.09; PSO 2.37, the only method not significantly different), also under the Gaussian wake at  $1^\circ$  (2.22 vs. 2.35); PSO’s narrow lead under the Gaussian wake with  $15^\circ$  bins (1.91 vs. 1.97;  $\bar{\tau} = 0.66$ ) is a property of the coarse rose. Beyond the top two the order changes ( $\bar{\tau} = 0.52$ ; same best method in 37% of the cases; MS-SLSQP moves from sixth to third). The case-mean difference PSO-VNS – PSO becomes  $-0.043$  pp (90% CI  $[-0.062, -0.025]$ ): a small but significant advantage, not equivalence at  $\pm 0.05$  pp; the Data Set II  $N \geq 10$  advantage remains (10 of 10 cases). The component-analysis conclusions hold, but the small SSA

gain over RS-VNS is no longer significant. The rankings are thus properties of the discretized benchmark.

The optimized layouts often lie on the  $4D$  constraint: 49% of the feasible final layouts also satisfy  $5D$  and 36%  $6D$  (4.1% and 0.1% for  $N = 11$ – $15$ ), so they do not transfer to larger spacings without re-optimization (Tables S40 and S41). The boundary rule also matters: an SSA that projects turbines radially onto the circle instead of clipping the coordinates attains higher mean objectives than the original SSA in all six representative cases ( $p \leq 0.0036$ ), so all methods use the same rule (Section II-C).

## X. DISCUSSION AND RECOMMENDATIONS

In this benchmark, which method appears best depends on the protocol as much as on the methods. An advantage over PSO is informative only if the PSO baseline is convergent: the old setting lies beyond the order-2 boundary, which marks approximately where this PSO starts to fail, and places PSO at position 5 of Table I instead of 1 (Section VI-A). An advantage of a hybrid's first phase is informative only if it beats random sampling in the farm at equal cost: square sampling is too weak a control, and against disc sampling only the PSO phase gains (Section VII); as the second phase is essentially one compass-search descent, this measures the value of a start for a local search. Non-significance is not equivalence, and equivalence depends on the level and the evaluation: PSO-VNS and PSO are equivalent at  $\pm 0.05$  pp across seeds and cases but not across farm clusters, nor under a  $1^\circ$  rose, where PSO-VNS has a small significant advantage (Sections VI-B and IX). From feasible starts, a single local search (VNS alone) ranks first (Section VIII-A). We recommend [23], [24]:

- 1) use convergent baseline settings (for PSO, (8) or (7)), state the bound and velocity handling, and verify the behavior with instrumented runs;
- 2) give all methods the same budget, counting the initial population and gradient calls, pair runs by seed, and rank feasibility first;
- 3) use multiplicity-corrected tests [22], and report threshold and cluster-level robustness;
- 4) test equivalence at a stated level (seeds, cases or clusters) and margin, and report the minimal margin [25], [26];
- 5) in ablations of hybrids, include the local method alone and an in-site random-sampling control of equal cost [57];
- 6) re-evaluate final layouts at a finer direction resolution, report this discretization sensitivity, and publish code and data.

## XI. LIMITATIONS

All results are for discretized roses ( $15^\circ$  benchmark bins,  $5^\circ$  on Horns Rev 1), which optimized layouts exploit, a top-hat Jensen wake, circular farms, a  $4D$  spacing and a prescribed  $N$ , with random starts and 6,030 evaluations unless stated. With  $1^\circ$  bins no optimized Horns Rev 1 layout exceeds the installed block; our Horns Rev 1 model takes  $C_T$  at the free-stream speed (wake loss  $-0.72$  pp relative to PyWake's NOJ model at identical bins). On IEA37, the best PSO-VNS

layouts at 30,030 evaluations have AEP gaps of  $-1.93\%$  (16 turbines) and  $-6.23\%$  (36 turbines) to the best feasible published layouts, radially projected onto the boundary. At 120,030 evaluations the leading methods lie within 0.17 pp, and at 6,030 the VNS phase is a truncated compass search. The pool lacks CMA-ES [58] and L-SHADE, DE is untuned, MS-SLSQP uses finite differences on a discontinuous objective, and the old PSO setting was run without velocity clamping, which largely restores its feasibility (Section VI-A). The margin, the  $N \geq 10$  split and the density pattern were stated after the primary analysis, and the p-values describe this benchmark of six clusters with nested  $N$ . Turbulence and loads (critical at  $4D$ ), wake steering, blockage, terrain, cables, electrical losses and costs are outside our scope; all AEP values are gross.

## XII. CONCLUSION

On the discretized Kusiak–Song benchmark with random starts and 6,030 evaluations, three findings emerge. First, the baseline configuration changes the conclusions: the old PSO setting, which lies beyond the order-2 boundary and rarely samples feasible layouts late in the run, puts PSO behind salp-swarm hybrids, whereas the constriction setting puts it ahead of every salp-swarm method at this budget. Second, salp-swarm first phases add nothing beyond random sampling in the farm (a gain above the margin is ruled out across seeds and cases; LX-SSA as published is worse), whereas a PSO phase gives the second phase, essentially one compass-search descent, a clearly better start. Third, PSO-VNS has the best average rank in every evaluation except the Gaussian wake with  $15^\circ$  bins, but its average advantage over a well-configured PSO is small: within  $\pm 0.05$  pp under the benchmark rose (across seeds and cases, not farm clusters) and  $-0.043$  pp under a  $1^\circ$  rose, with case-level differences in both directions. PSO-VNS reaches feasibility more reliably on Horns Rev 1 (30/30 vs. 19/30 runs, McNemar  $p = 9.8 \times 10^{-4}$ ); it is a consistent reference method, not a superior optimizer. Future work will address larger budgets and farms, finer roses, calibrated wake models, stronger baselines and economic objectives.

## ACKNOWLEDGMENT

[TBD: acknowledgment, or delete this section]

## DATA AVAILABILITY

The source code of all methods and of the benchmark evaluator, the per-run results of all studies (including the runs with the old PSO setting) and the analysis scripts are openly available at [TBD: Zenodo DOI]. Each run record gives the method, case, seed, final objective (AEP for Horns Rev 1), feasibility, final turbine coordinates and convergence curve, and the budget and initialization where these vary. The file `analysis/README_reproduce.md` lists the script that produces each table, figure and number of this paper and of its supplementary material, and `analysis/requirements.txt` pins the software versions. The Horns Rev 1 and IEA37 input data are taken from [53], [54]. The supplementary material gives the model details, the pseudocode of SSA and LX-SSA, the wind data,

the per-case results, the coordinates of the best layouts and further figures.

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**Prince Solanki** [TBD: biography: degrees, institutions, research interests]

**Pulkit Dwivedi** [TBD: biography]

**Vanita Garg** [TBD: biography]

**Vinay Shukla** [TBD: biography]